Resonance & tuned mass damper concept

EMAE 285 · independently proposed experiment

Experiment overview

Designed a one-degree-of-freedom flexible structure with a 100 g primary mass and a proposed 15 g secondary tuned mass damper. Planned a frequency sweep to compare vibration response before and after adding the damper.

Variables and instrumentation

Proposed mass, stiffness, damping, forcing frequency

Analysis and presentation

Predicted resonance curves; planned comparison (not measured results)

Engineering relevance

Demonstrates experimental initiative and the ability to turn vibration theory into a test plan.

Scope and attribution

Course laboratory / group work unless noted. The resonance experiment is a proposed design and has no claimed measured outcomes.

Portfolio case-study brief based on coursework and user-provided descriptions. This is NOT a copy of the original laboratory report. Diagrams on the website are explanatory illustrations, not photographs or original experimental plots.

Original source (may require Google Drive permission):
https://docs.google.com/document/d/1dijlXiE-iIi2WLW5KhdR3EoZVeio71RSIHyj4p97_9I/edit